

# A Neutral Interlingua for LLM Hidden States: Micro-Translators, Auditable Vector Channels, and the Limits of the Shared Representational Core

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## Abstract

Multi-agent LLM systems communicate through natural language, paying a decode/re-encode tax in latency and information loss, and—as recent work on subliminal learning suggests—potentially carrying hidden, model-specific signals. We present a protocol in which frozen, heterogeneous LLMs exchange hidden states directly through a jointly learned *neutral hub*: for  $N$  models,  $2N$  small encoder/decoder “micro-translators” ( $\sim 1\text{--}5\text{M}$  parameters each) replace  $N(N-1)$  pairwise bridges. On consumer hardware, across three independent model families (Llama-3.2-1B, Qwen2.5-0.5B, SmolLM2-360M) and a 10,000-concept bilingual benchmark, we show: (i) translated states preserve identity (up to 88% top-1 retrieval over a 10k pool) and geometry (Procrustes rank correlation  $\rho = 0.995$ ); (ii) after a tuned-lens-style KL-distillation phase, injected translations are functionally near-indistinguishable from the receiver’s own states (KL ratio 0.17 vs. a random control at concept level; 0.016 at sentence level); (iii) a BatchTopK sparse-autoencoder bridge sustains  $8\times$  compression at a 3.5-point retrieval cost, putting a message at 64–128 bytes; (iv) the vector channel is  $\sim 900\times$  faster than a text round-trip; (v) adding a third model family *improves* existing routes (+15 points), suggesting the shared space is regularized rather than crowded by new members. We also report a measured limit: relative translation fidelity is language-dependent (English 1.01—lossless; Turkish 0.39) even though every model encodes Turkish concepts near-perfectly in isolation, indicating that the “Platonic” shared core is thick where training distributions overlap and thin elsewhere. Code and benchmarks: <https://github.com/OrhanKorkmaz/ucm-mct>.

## 1 Introduction

LLM agents today cooperate by generating text that other agents re-read. This channel is expensive—every message pays a full decode and re-encode cycle—and lossy: graded uncertainty, partial inferences, and other non-verbalized state do not survive serialization. Prior work has begun to bypass the token bottleneck: activation-level communication between agents [1], latent-only inter-agent channels [2], state-delta trajectories [3], and unsupervised embedding-space translation [4]. These efforts are mostly pairwise: each pair of models needs its own bridge.

We ask a stronger question: can heterogeneous, frozen LLMs share a single *neutral* representation space that none of them owns—an interlingua for hidden states—so that joining the network costs one encoder and one decoder, independent of network size? We call the resulting bridges *micro cognitive translators* (MCT).

Our contributions: (1) a complete, reproducible protocol (extraction, CKA-based layer pairing, standardization, translator and hub training, injection, compression) validated end-to-end on consumer hardware; (2) the observation that *functional* compatibility does not follow from vector-space proximity, and a distillation recipe that reliably closes the gap at both concept and sentence level; (3) a positive scaling result for the neutral hub (a third family improves existing routes); and (4) a measured, language-dependent limit of cross-model representational overlap that constrains any interlingua approach.

## 2 Method

**Micro cognitive translators.** An MCT maps source states  $x \in \mathbb{R}^{d_A}$  to target states in  $\mathbb{R}^{d_B}$ :

$$y = \mu_{\text{out}} + sW(x - \mu_{\text{in}}) + g \cdot \text{MLP}(x - \mu_{\text{in}}),$$

with  $W$  initialized by orthogonal Procrustes on training pairs and kept semi-orthogonal by a soft penalty  $\|W^\top W - I\|^2$  (hard orthogonality parametrizations were numerically unstable in our setting);  $g$  gates a residual MLP that starts at zero. All training operates on per-dimension z-standardized activations—indispensable in practice because massive-activation dimensions otherwise dominate every objective.

**Neutral hub.** Each model  $m$  contributes spokes (selected layers) with an encoder  $E_m$  into a shared  $d_h$ -dimensional space and a decoder  $D_m$  back. The hub is trained jointly with symmetric InfoNCE across all spoke pairs, reconstruction, cross-translation anchors, and vec2vec-style cycle consistency ( $A \rightarrow \text{hub} \rightarrow B \rightarrow \text{hub} \rightarrow A$ ). No pairwise translators are trained; every route goes through the hub.

**Functional alignment by distillation.** Vector-space training alone produces translations that are *close* (cosine  $\approx 0.73$ ) yet *functionally destructive* when injected: the residual error is out-of-distribution for the receiver’s downstream layers. We therefore add a tuned-lens-style phase: the translated vector is injected into the frozen receiver at the target layer (single position, norm-calibrated), and the KL divergence between the resulting next-token distribution and the receiver’s clean distribution is minimized, with gradients flowing only into the translator. This phase moved concept-level injection agreement from 27% to 81%, and sentence-level agreement from 27% to 83% (KL ratio 0.016).

**Compression.** Messages are hub vectors passed through a BatchTopK sparse autoencoder with an auxiliary dead-feature revival loss; only the (*index*, *value*) pairs of the  $k$  active features travel.

## 3 Experimental setup

Models: Llama-3.2-1B (16 layers,  $d = 2048$ ), Qwen2.5-0.5B (24 layers,  $d = 896$ ), SmolLM2-360M (32 layers,  $d = 960$ ); all frozen, bfloat16, Apple M2 (16 GB). Benchmark: 10,000 frequency-ranked concepts (8,000 English, 2,000 Turkish), 80/20 split; contextual last-token states under concept-final templates; layer pairs selected by linear CKA within relative depth bands. Injection protocol: three conditions (clean / translated / random real vector), single-position replacement with norm calibration.

## 4 Results

| Claim              | Metric                                                  | Result                                 |
|--------------------|---------------------------------------------------------|----------------------------------------|
| Geometry preserved | Procrustes rank corr.                                   | $\rho = 0.995$                         |
| Identity preserved | top-1 retrieval (10k pool)                              | up to 88%                              |
| Function preserved | KL ratio vs. random                                     | 0.17 (concept), 0.016 (sentence)       |
| Compression        | top-1 drop @ $8\times$                                  | $-3.5$ pts, 0 dead features            |
| Depth translation  | cosine ( $L_2 \rightarrow \text{hub} \rightarrow L_3$ ) | 0.778                                  |
| Channel latency    | ms/message                                              | 0.32 vs. 286 text ( $\sim 900\times$ ) |
| Third family joins | effect on existing routes                               | +15 pts                                |

Table 1: Headline results. Every row maps to a JSON artifact in the public repository.

**Scaling behavior.** With two families, the semantic route Llama $\rightarrow$ Qwen reached 72.5% top-1 through the hub. Adding SmolLM2-360M (two new translators only) raised it to 87.7%, and all six cross-family routes landed at 79–88%. New members appear to act as regularizers of the shared space, analogously to multilingual training.

**The relative-fidelity limit.** We measure each model’s *self-consistency ceiling* (retrieving a concept encoded under one template from a pool encoded under another;  $\geq 0.98$  for both languages in all models) and compare cross-model translation against it. English:  $0.988/0.979 = 1.01$ —the channel is lossless. Turkish:  $0.387/0.988 = 0.39$ . Both libraries know Turkish; their Turkish *geometries* do not align. Functional distillation at sentence level narrows the Turkish gap substantially (72% vs. 86% agreement), indicating that functional alignment is easier than identity alignment where the shared core is thin.

**Robustness.** Simulated representation drift (mixing 40% of an adjacent layer’s geometry plus noise) cost only  $\sim 1$  point through the hub; a 500-concept canary set with closed-form Procrustes repair restores performance when degradation does occur, and should be triggered only below a degradation threshold—premature recalibration on small canaries hurts.

## 5 Related work

Model stitching [6, 7]; activation communication between agents [1]; latent-only communication [2]; state-delta encoding [3]; unsupervised embedding translation [4]; tuned lens [5]; crosscoders and model diffing [8]; subliminal learning [9]; introspective detection of injected concepts [10]. Our work differs in the N-model neutral-hub formulation, the cross-model depth translation, the distillation-based functional alignment recipe, and the measured language-dependence of the shared core.

## 6 Limitations and safety

All models are  $\leq 1\text{B}$  parameters; scale generalization is untested. Sentence-level results use template-generated contexts, not free text. The receiver could not name injected concepts (0% at 0.5B), consistent with introspection emerging at scale. A vector channel is a new attack surface; we propose (and will evaluate) an SAE-based firewall that passes shared semantic features and drops

model-exclusive components—the same mechanism that makes the channel auditable makes it defensible.

## 7 Conclusion

A neutral interlingua for LLM hidden states is buildable today, on a laptop: identity, geometry, function, compression, and speed all survive translation, and the hub improves as families join. The measured thinness of the shared core outside high-resource language regions is, to our knowledge, the first quantitative boundary drawn for this family of approaches—and a concrete target for making machine communication genuinely universal.

## References

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